lower word

Division:

9A23H ÷ 00FFH

Quotient = 0911H

Remainder = 0002H.

Ex No. 2

1) AND OR, XOR and Nibble masking

given : X = 5CH & Y = 3AH

case 1: X = 5CH, Y = 3AH

5CH - 01011100

3AH = 0011 1010

case 2: X = E7H, Y = 81H

E7H = 1110 0111

81H = 1000 0001

Binary = 6DH = 0110 1101

One's complement:

1001 0010 (92H)

Two's complement.

```
  92H
+ 01H
------
  93H
```

Verification:

```
  6DH
+ 93H
------
 100H
```

The 8-bit result is 00H, with carry = 1

16 bit number = 4F20H

One's complement: 4

B0DFH

4F20H = 0100 1111 0010 0000

B0DFH = 1011 0000 1101 1111

Two's complement:

```
  B0DFH
+ 0001H
--------
  B0E0H
```

```
  4F20H
+ B0E0H
--------
 10000H
```

The 16-bit result is 0000H, with carry = 1

80H = 1000 0000

1's of 80H = 0111 1111 = 7FH

7FH + 1 = 80H

3) Input = 93H

Binary = 93H = 1001 0011

operation	manual calculation	Result
Set bits 2,5	93H OR 24H	B7H
Clear bits 0,7	93H AND 7EH	12H
Toggle bit 4	93H XOR 10H	83H
Test bit 3	93H AND 08H = 00H	Bit 3=0

Input = 6CH

6CH = 0110 1100

operation	Manual	Result
set bit 2,5	6CH OR 24H	6CH
clear bit 0,7	6CH and 7EH	6CH
Toggle bit	6CH XOR 10H	7CH
Test bit	6CH AND 08H = 08H	Bit 3=1

4) Enter B4H = 10110100

Logical shift left	68H	1
Logical shift right	5BH	0
Arithmetic shift right	EDH	0
RLC	69H	1
RRC	5BH	0

5)

Input	Binary	Number of 1s	parity	output
7EH	01111110	6	Even	00H
2BH	00101011	5	odd	01H